AI 101 for Mechanical Engineers · Module 12

Sequence architectures are different mechanisms

Lesson 12.2

What you will learn

> Compare recurrent and attention-based sequence mechanisms, explain why attention does not prove causal importance, and separate issue-time information from future measured data.

Lesson overview

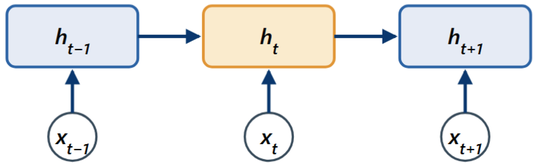
Module 12 · Lesson 2

Sequence architectures are different mechanisms

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1 Recurrent models carry state

An RNN carries a hidden state from one sequence step to the next.



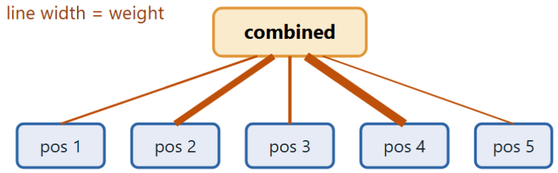
$$h_t = f(W_x x_t + W_h h_{t-1} + b)$$

LSTM: a recurrent architecture with mechanisms to retain and update information.

A hidden state does not guarantee every useful detail is retained.

2 Transformers use attention

Attention weights and combines representations across sequence positions, without a recurrent state.



When order matters, supply a positional representation.

Attention weights do not prove a physical cause.

Compare with a fair baseline, such as a simple lagged-feature model.

3 Use inputs available at issue time

Declare the forecast issue time, target time and horizon.

✓ **Earlier measured observations**
Use if already available.

! **Previously issued forecasts**
May be available; record their uncertainty.

✕ **Future measured values**
Unavailable at issue time. Exclude them.

Tomorrow's measured weather is not an input to a forecast issued today.

Inspect timestamps and missing values first; forward fill or interpolation is not automatically safe.

Key takeaway **Recurrence and attention are different mechanisms. Every operational input must exist at the declared issue time.**

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An RNN carries a hidden state forward

- The state provides a mechanism to retain earlier information [4]
- An LSTM has mechanisms to retain and update information
- These describe information handling, not engineering accuracy

A simplified RNN update

$$h_t = f(W_x x_t + W_h h_(t\text{-}1) + b)$$

- x_t: current input; h_(t-1): previous state; h_t: updated state
- Earlier information can be carried forward
- No guarantee every useful detail is retained over long sequences

Recurrence and attention are different mechanisms

RNN and LSTM

- Carry a hidden state step to step [4]

Transformer

- Attention combines representations across positions [63]
- Needs positional representation when order matters

Different mechanisms [63], [66]; choose by data, constraints, evidence, fair baseline (e.g. lagged-feature model).

Attention weights are not explanations or causes

- Weights describe contributions to the model computation
- They are not human attention
- Alone, not a reliable explanation of a prediction [64], [65]
- They do not prove a variable is a physical cause [26]

State the forecast issue time

- State the issue time and the target time or horizon
- Use only information available at issue time [18, Sec. 7.6]
- These are called ex-ante forecasts [18, Sec. 7.6]
- Inputs must physically exist at the declared issue time

What is available at issue time?

Input	Operational use
Observed load before issue	May be available
Earlier published weather forecast	May be available; record its uncertainty
Tomorrow’s measured weather	Unavailable; must not be used

> Using future measured inputs evaluates a different information scenario.

Inspect missing signal values first

1. Gaps: sensor dropout, communication loss, calibration work
2. Inspect timestamps and missingness
3. Choose a treatment fitting sampling and gap length
4. Forward fill or interpolation is not automatically safe
5. Record any imputation or excluded interval

CHECK YOUR UNDERSTANDING

How do recurrence and attention differ, and does either show which input causes the output?

Answer before moving on.

CHECK YOUR UNDERSTANDING

How do recurrence and attention differ, and does either show which input causes the output?

> Recurrence carries a hidden state through steps; attention combines representations across positions. Both need an appropriate representation of time or order; neither alone establishes causation.

Key takeaways

- RNNs and LSTMs carry hidden states; transformers use attention
- Attention is computational weighting, not proof of causal importance
- Define issue time, target time and forecast horizon
- Future measured data cannot be operational inputs

References (1/2)

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[26] J. Pearl, Causality: Models, Reasoning, and Inference, 2nd ed. Cambridge, U.K.: Cambridge Univ. Press, 2009. doi: 10.1017/CBO9780511803161

[63] A. Vaswani et al., “Attention is all you need,” in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 30, 2017, pp. 5998–6008.

Numbers match the course reference list. The written lesson is the authoritative text.

References (2/2)

[64] S. Jain and B. C. Wallace, “Attention is not explanation,” in Proc. 2019 Conf. North Amer. Chapter Assoc. Comput. Linguistics: Human Lang. Technol. (NAACL-HLT), 2019, pp. 3543–3556. doi: 10.18653/v1/N19-1357

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[66] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” Neural Comput., vol. 9, no. 8, pp. 1735–1780, 1997. doi: 10.1162/neco.1997.9.8.1735

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